

— JCSSE 2026 ▪ STREAMING ML ▪ EXPLAINABLE AI

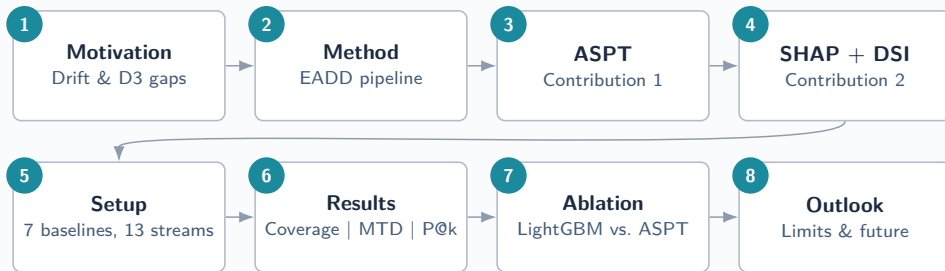
Enhanced Adversarial Drift Detection

for MLOps Feature Monitoring

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Today's roadmap



EADD answers **whether**, **when**, **which**, and **how severe** — in one pipeline.

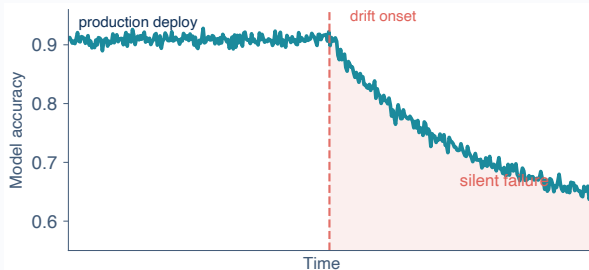
01

— SECTION 01

The Problem

Why production ML detectors miss the mark

Production ML models fail silently



Distribution drift silently degrades models. Existing detectors raise a binary alarm; practitioners need:

Which features?

How severe?

Real or noise?

D3

Gözüaçık+ CIKM 2019

Most robust unsupervised detector (Lukats+ 2024)—yet has **three gaps**:

- 1 No formal significance test**
→ false alarms on autocorrelated streams
- 2 No feature attribution**
→ “detect-and-discard” classifier
- 3 Linear classifier**
→ misses non-linear / incremental drift

Capability matrix | the integration gap

WHETHER

Binary drift alarm

WHICH

Feature attribution

HOW SEVERE

Severity quantification

Approach	Whether	Which	Severity	Streaming
DDM / EDDM	✓	×	×	✓
KS / KSWIN	✓	~	×	✓
D3	✓	×	×	✓
Vertex AI / Clarify	✓	✓	~	×
TRIPODD / HDPD	✓	✓	×	×
EADD (this work)	✓	✓	✓	✓

Three contributions, one pipeline

1

ASPT

Adaptive Sequential Permutation Test

Formal permutation test gates drift confirmation.

- stat rigor
- 90–95% fewer perms

Inspired by Gandy 2009.

Confirms at $i=20$ vs. $B_{\max}=199$.

2

SHAP + DSI

Attribution & Severity

TreeSHAP on the C2ST classifier; entropy-normalised severity.

- which features
- how severe

$$\text{DSI} = \text{AUC}(1 - H_{\text{norm}}(\mathbf{s}))$$

3

Benchmark

End-to-End Evaluation

EADD vs. 7 unsupervised detectors on synthetic + 13 real-world streams.

- 4/4 coverage
- 0 false alarms

$P@k=R@k=1.0$ (univariate).

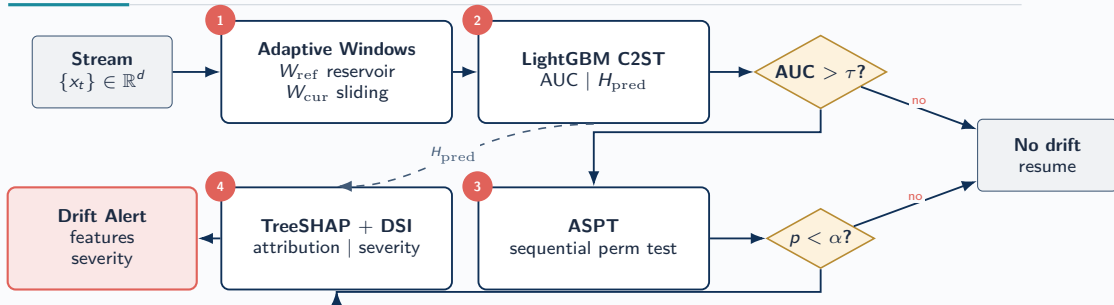
02

— SECTION 02

The EADD Pipeline

Four-step streaming detector

End-to-end detection pipeline



Two-stage gate $\Rightarrow$ Type I error bound: $\Pr(\text{AUC} > \tau \mid H_0) \cdot \alpha \leq \alpha$.

ASPT — Adaptive Sequential Permutation Test

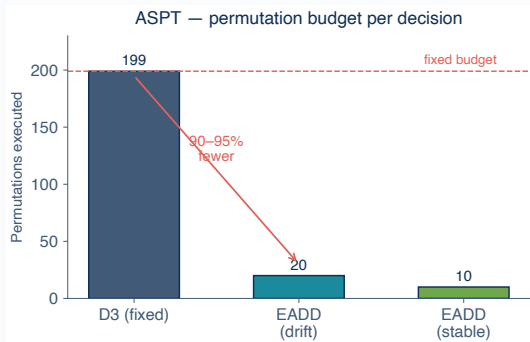
Problem. D3's fixed AUC threshold $\rightarrow$ false alarms on autocorrelated streams.

Idea. Permute labels iteratively; stop early once evidence is decisive.

Early reject H_0 : $c_i=0, i \geq B_{\min}, 1/(i+1) < \alpha$
 $\Rightarrow$ confirm drift, $p = 1/(i+1)$

Early accept H_0 : $c_i/i > 2\alpha, i \geq B_{\min}$
 $\Rightarrow$ no drift, $p = (c_i+1)/(i+1)$

Terminal: $p = (c_{B_{\max}}+1)/(B_{\max}+1)$



90-95% fewer permutations | α preserved

SHAP attribution + Drift Severity Index

TreeSHAP on the C2ST classifier (which D3 discards).

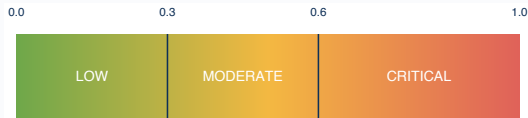
$$\bar{\phi}_j = \frac{1}{n} \sum_i |\phi_j^{(i)}|, \quad s_j = \bar{\phi}_j / \sum_k \bar{\phi}_k$$

Drift Severity Index.

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s}))$$

$H_{\text{norm}}(\mathbf{s}) = - \sum_j s_j \log_2 s_j / \log_2 d \in [0, 1]$ —
dimensionality-invariant.

Severity gradient & prescription tiers



Tier	Recommended action
Low (<0.3)	Diffuse shift → scheduled retrain
Moderate (0.3–0.6)	Subset drift → audit shared source
Critical (>0.6)	Concentrated → audit pipeline

Heuristic tiers; require domain calibration.

03

— SECTION 03

Experiments & Results

What does EADD actually do?

Experimental setup

EXP 1 Synthetic

$n=5,000$, $d=5$

4 drift patterns:

abrupt, gradual, incremental,
recurring

*Also: 4 stable streams for false-
alarm robustness*

EXP 2 Real-world

13 streams (Lukats+ 2024)

4 annotated INSECTS

9 unannotated

Metrics: Precision, Recall, MTD

EXP 3 SHAP accuracy

$n=10,000$, $d=10$

drift @ $t=5,000$

univariate / subset / multivariate

Metrics: $P@k$, $R@k$, DSI

Baselines: D3, BNDD, CSDDM, IBDD, OCDD, SPL, UDetect. **Config:** Light-GBM (50 trees), $N_{\text{ref}}=500$, $N_{\text{cur}}=200$, $M=50$, $B_{\text{max}}=199$, $B_{\text{min}}=10$, $\alpha=0.05$, $\tau=0.7$.

Headline results

4/4

synthetic
coverage

0

false alarms
on stable streams

8.3×

faster MTD
vs. D3 (incremental)

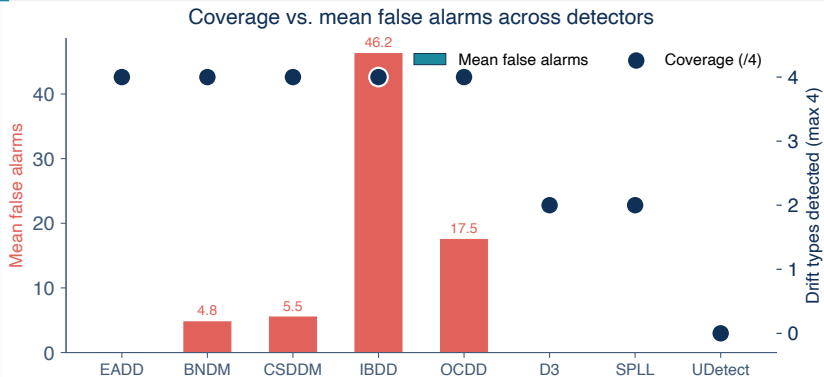
1.00

P@k=R@k
(univariate SHAP)

EADD is the only detector that achieves both full drift-type coverage **and** zero false alarms, while answering *which* and *how severe* — all in one streaming pipeline.

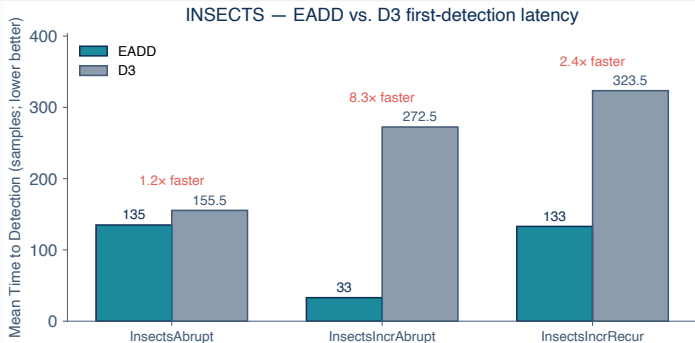
Mann–Whitney U vs. D3 on autocorrelated streams: $p = 0.0101$ ($\tau=0.6$). Source data: [experiments/results/](#).

Synthetic coverage & false alarms



EADD: 4/4 coverage AND 0 false alarms. BNDD/CSDDM/IBDD/OCDD also hit 4/4 but pay heavy FA cost; D3/SPL keep 0 FA only by missing half the drift types.

INSECTS — faster detection on incremental drift

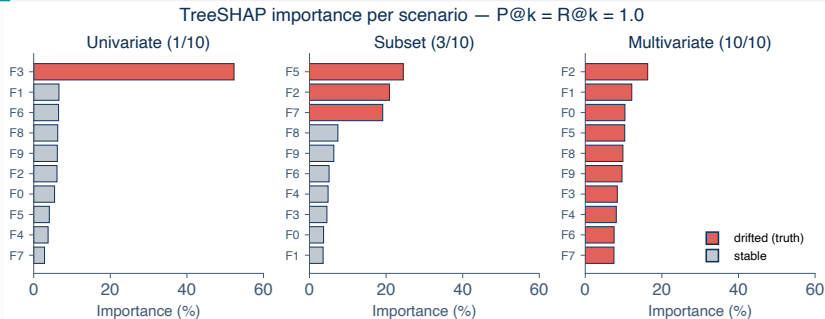


8.3x faster
InsectsIncrAbrupt — 33 vs. 272.5

2.4x faster
InsectsIncrRecur — 133 vs. 323.5

1.15x faster
InsectsAbrupt — 135 vs. 155.5

SHAP attribution — $P@k = R@k = 1.0$

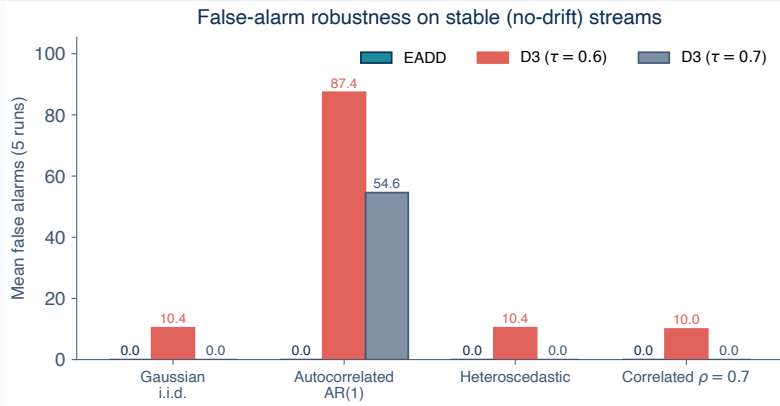


Scenario	AUC	Top feat.	DSI
Univariate (1/10)	0.784	F3 (52.3%)	0.193
Subset (3/10)	0.809	F5 (24.5%)	0.096
Multivariate (10/10)	0.806	F2 (14.6%)	0.016

DSI orders by concentration:

univariate > subset > multivariate (*consistent with H_{norm}*)

False-alarm robustness on stable streams



0 false alarms on all 4 stable stream types. Mann–Whitney U vs. D3 ($\tau=0.6$): $p = 0.0101$. ASPT suppresses 54.6 spurious D3 detections on autocorrelated AR(1) at $\tau=0.7$.

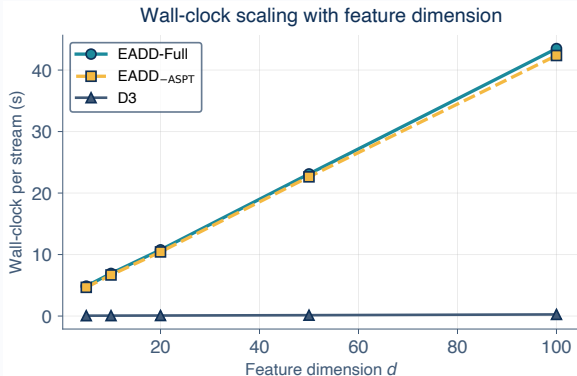
04

— SECTION 04

Ablation & Cost

Where does the gain come from?

Where does the gain come from?



Decomposing the gain

Coverage 4/4 vs. D3's 2/4

comes from **LightGBM** (non-linear C2ST).

→ EADD-ASPT also reaches 4/4.

0 false alarms on AR(1)

comes from **ASPT**.

→ suppresses 54.6 spurious D3 detections.

Cost trade-off

~50–160× D3 wall-clock — dominated by LightGBM, not ASPT (< 5%).

05

— SECTION 05

Limits & Outlook

What EADD cannot answer—yet

Limitations & future work

Limitations

- Wall-clock $\sim 50\text{--}160\times$ D3—LightGBM retrains every $M=50$ steps.
- **Multiple testing over time:** ~ 600 tests/30K samples; Bonferroni $\alpha' \approx 8.3 \times 10^{-5}$ below ASPT's $1/(B_{\max}+1)=0.005$ floor.
Zero-FA result empirical, not theoretically guaranteed.
- DSI tier boundaries heuristic; never triggered above LOW in experiments.
- SHAP attributions correlational, not causal.
- No isolation of reservoir sampling vs. window config contributions.

Future work

- Anytime-valid **e-values** for streaming FWER control.
- **Causal** attribution (Budhathoki+ AISTATS 2021).
- Multi-domain **DSI calibration** to ground tier boundaries empirically.
- Adversarial robustness of discriminative drift detectors.
- Per-feature operating curves under correlation $\rightarrow$ feature-group SHAP.

Open invitation. EADD is a starting point—each gap above is a separate research thread.

Conclusion

EADD answers **WHETHER, WHEN, WHICH**, and **HOW SEVERE** — in one pipeline.

4/4

coverage

0

false alarms

8.3×

faster MTD

1.00

SHAP P@k

What we built

- ASPT + H_{pred} — rigorous confirmation, 90–95% perm savings.
- SHAP + DSI — feature-level diagnosis + severity.
- Benchmark vs. 7 unsupervised detectors.

What it gives MLOps

- Reliable alarms on autocorrelated production streams.
- Actionable diagnoses, not just binary flags.
- Severity-tiered prescriptions for response.



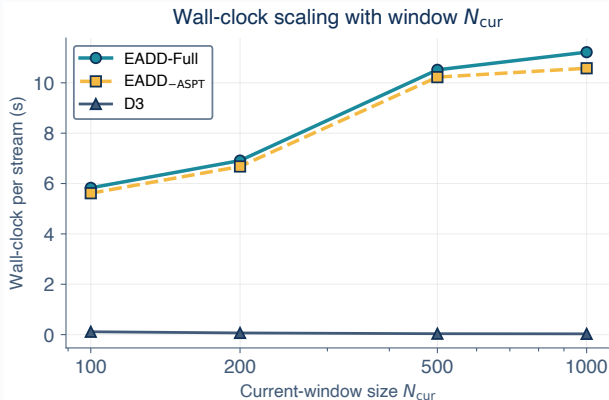
Thank you.

Questions & discussion welcome.

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Code + paper available at the project repository.

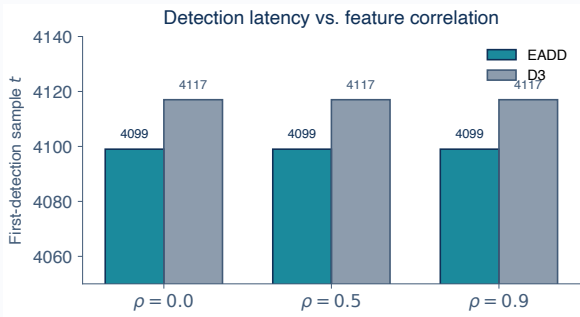
Backup | runtime scaling vs. window size



- $N_{\text{cur}}=200 \rightarrow 1000$: EADD time grows only $1.6\times$.
- LightGBM cost sub-linear in N , near-linear in d .
- ASPT overhead $< 5\%$ across window sizes.

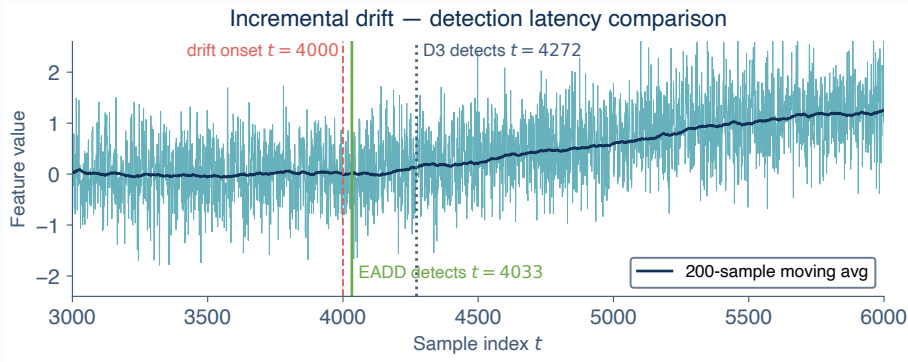
Takeaway. Increasing N_{cur} for tighter detection incurs sub-linear cost; bottleneck is per-step LightGBM training.

Backup | correlated features ($d=10$)



- Cholesky-mixed Gaussian; $+2\sigma$ shift in 3 features at $t=4000$.
- EADD detection robust to correlation; stable correlated streams give **0 FA**.
- SHAP under correlation $\rightarrow$ interpret at **feature-group** granularity.

Backup | incremental-drift timeline



Illustrative INSECTS-style trace; EADD detects within ~ 33 samples vs. D3's ~ 272.5 .